Sharp Hölder bounds for the Fourier spectrum: the compact and noncompact answers differ

Candidate full solution to Question 1.4 of arXiv:2210.07019

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Status: candidate full solution, likely valid. Let $t = \dim_{\mathrm{F}} \mu < \infty$. The general upper bound in Theorem 1.3 of Fraser,

$$\dim_{\mathrm{F}}^{\theta} \mu \le t + d\left(1 + \frac{t}{2\alpha}\right)\theta, \tag{A}$$

is sharp for every $d \ge 1$, $0 < \alpha \le 1$, and $t \ge 0$: a finite positive measure with an α-Hölder Fourier transform attains equality for every $0 \le \theta \le 1$. In contrast, compact support implies the strictly stronger exact bound

$$\dim_{\mathrm{F}}^{\theta} \mu \le t + d\theta. \tag{B}$$

For every $t \ge 0$, a compactly supported positive measure attains equality in (B) for every θ. Thus (A) is not sharp in the compactly supported class when $t > 0$, and (B) is the optimal replacement.

1 The exact question

For a finite Borel measure μ on $\mathbb{R}^d$, Fraser defines

$$\mathcal{J}_{s,\theta}(\mu)^{1/\theta} = \int_{\mathbb{R}^d} |\widehat{\mu}(\xi)|^{2/\theta} |\xi|^{s/\theta - d}\, d\xi \qquad (0 < \theta \le 1)$$

and $\dim_{\mathrm{F}}^{\theta} \mu = \sup\{s \ge 0 : \mathcal{J}_{s,\theta}(\mu) < \infty\}$. At $\theta = 0$ this is the usual Fourier dimension

$$\dim_{\mathrm{F}} \mu = \sup\{s \ge 0 : |\widehat{\mu}(\xi)| \lesssim |\xi|^{-s/2}\}.$$

Theorem 1.3 proves (A) whenever $\widehat{\mu}$ is α-Hölder. Question 1.4 asks whether this is sharp, especially when μ is compactly supported and $\dim_{\mathrm{F}} \mu > 0$ [1]. The complete source page is reproduced below.

Page 7 J. M. Fraser

Since μ is α-Hölder, there exists $c = c(\mu) \in (0,1)$ such that

$$|\widehat{\mu}(z)| \geqslant |z_k|^{-t/2}/2$$

for all $z \in B(z_k, c|z_k|^{-t/(2\alpha)})$. By passing to a subsequence if necessary we may assume that the balls $B(z_k, c|z_k|^{-t/(2\alpha)})$ are pairwise disjoint. Therefore

$$\begin{aligned}\mathcal{J}_{s,\theta}(\mu)^{1/\theta} &\geqslant \sum_k \int_{B(z_k,c|z_k|^{-t/(2\alpha)})} |\widehat{\mu}(z)|^{2/\theta}|z|^{s/\theta-d}\,dz \\ &\gtrsim \sum_k |z_k|^{-td/(2\alpha)}|z_k|^{-t/\theta}|z_k|^{s/\theta-d} = \infty\end{aligned}$$

whenever $s \geqslant d\theta + t + td\theta/(2\alpha)$. This proves

$$\dim_{\mathrm{F}}^{\theta} \mu \leqslant \dim_{\mathrm{F}} \mu + d\left(1 + \frac{\dim_{\mathrm{F}} \mu}{2\alpha}\right)\theta$$

as required. The final conclusion that $\dim_{\mathrm{F}}^{\theta} \mu$ is Lipschitz continuous on $[0,1]$ follows immediately from Lipschitz continuity at $\theta = 0$ together with the fact that $\dim_{\mathrm{F}}^{\theta} \mu$ is non-decreasing and concave, see Theorem 1.1. $\square$

Later we provide an example showing that the bounds from Theorem 1.3 are sharp in the case $\dim_{\mathrm{F}} \mu = 0$ and $\alpha = 1$, see Corollary 3.3.

Question 1.4. *Are the bounds from Theorem 1.3 sharp? Are they sharp when μ is compactly supported and $\dim_{\mathrm{F}} \mu > 0$?*

One benefit of Theorem 1.3 is the demonstration that positive Fourier dimension of a measure can always be observed by averaging (in the case when $\widehat{\mu}$ is Hölder). That is, provided $\widehat{\mu}$ is Hölder, $\dim_{\mathrm{F}} \mu > 0$ if and only if $\dim_{\mathrm{F}}^{\theta} \mu > d\theta$ for some $\theta > 0$. This is potentially useful since positive Fourier dimension requires uniform estimates on $|\widehat{\mu}|$ which are *a priori* harder to obtain than estimates on the 'averages' $\mathcal{J}_{s,\theta}(\mu)$.

Using Theorem 1.3 we immediately get continuity of the Fourier spectrum for compact sets. However, using a trick inspired by [EPS15, Lemma 1] we can upgrade this to continuity of the Fourier spectrum for *all* sets.

Theorem 1.5. *If $X \subseteq \mathbb{R}^d$ is a non-empty set, then*

$$\dim_{\mathrm{F}}^{\theta} X \leqslant \dim_{\mathrm{F}} X + d\left(1 + \frac{\dim_{\mathrm{F}} X}{2}\right)\theta$$

for all $\theta \in [0,1]$. In particular, $\dim_{\mathrm{F}}^{\theta} X$ is Lipschitz continuous on the whole range $[0,1]$ with $\dim_{\mathrm{F}}^{0} X = \dim_{\mathrm{F}} X$ and, if X is Borel, $\dim_{\mathrm{F}}^{1} X = \dim_{\mathrm{H}} X$.

Proof. Let μ be a finite Borel measure supported by X. Let $f : \mathbb{R}^d \to [0,\infty)$ be a smooth function with compact support such that the Borel measure ν defined by $d\nu = f\,d\mu$ satisfies $\nu(X) > 0$. Then ν is supported on a compact subset of X. We claim that

$$\dim_{\mathrm{F}}^{\theta} \nu \geqslant \dim_{\mathrm{F}}^{\theta} \mu$$

for all $\theta \in [0,1]$. Together with Theorem 1.3 and Lemma 1.2, this claim proves the result. Since f is smooth and has compact support, it holds that for all integers $n \geqslant 1$ $|\widehat{f}(t)| \lesssim_n |t|^{-n}$ for $|t| \geqslant 1$. In particular, $\widehat{f}$ and f are L^1 functions. Therefore, by the Fourier inversion formula,

$$\widehat{\nu}(z) = \int_{\mathbb{R}^d} \widehat{\mu}(z-t)\widehat{f}(t)\,dt. \tag{1.1}$$

2 Compact support forces a better upper bound

Theorem 1 (compact-support bound)**.** *Let μ be a compactly supported finite Borel measure on $\mathbb{R}^d$ and let $t = \dim_{\mathrm{F}} \mu < \infty$. Then, for every $0 \le \theta \le 1$,*

$$\dim_{\mathrm{F}}^{\theta} \mu \le t + d\theta.$$

Proof. The claim at $\theta = 0$ is the definition, so fix $\theta > 0$. First suppose $t > 0$, and choose $0 \le v < t < u$. The definition of Fourier dimension gives

$$|\widehat{\mu}(\xi)| \le C_v(1 + |\xi|)^{-v/2}. \tag{1}$$

Choose $\chi \in C_c^{\infty}(\mathbb{R}^d)$ equal to one on a neighbourhood of $\operatorname{spt} \mu$. For $j = 1, \ldots, d$,

$$\partial_j \widehat{\mu} = -2\pi i\, \widehat{x_j \mu} = -2\pi i\big(\widehat{x_j \chi} * \widehat{\mu}\big).$$

Convolution with a Schwartz function preserves every polynomial decay estimate, so (1) implies

$$|\nabla\widehat{\mu}(\xi)| \le C_v'(1+|\xi|)^{-v/2}. \tag{2}$$

Since $u > t$, there is a sequence ξ_k, with $R_k = |\xi_k| \to \infty$, such that

$$|\widehat{\mu}(\xi_k)| \ge R_k^{-u/2}.$$

The mean-value theorem and (2) show that, for a fixed small $c > 0$,

$$|\widehat{\mu}(\xi)| \ge \tfrac{1}{2}R_k^{-u/2} \quad \text{when} \quad |\xi - \xi_k| \le cR_k^{-(u-v)/2}. \tag{3}$$

Pass to a subsequence so that these balls are disjoint. Their contribution to the energy is bounded below by a constant times

$$\sum_k R_k^{-u/\theta} R_k^{s/\theta - d} R_k^{-d(u-v)/2}.$$

Every summand is bounded below away from zero, and hence the energy diverges, whenever

$$s \ge u + d\theta + \frac{d\theta}{2}(u-v).$$

Let $u \downarrow t$ and $v \uparrow t$ to obtain $\dim_{\mathrm{F}}^{\theta}\mu \le t + d\theta$. If $t = 0$, the same proof uses $v = 0$: boundedness of $\widehat{\mu}$, together with the cutoff identity, gives a bounded gradient. Letting $u \downarrow 0$ gives the same conclusion. □

The gain over Theorem 1.3 is structural. Hölder continuity propagates a peak of height $R^{-u/2}$ only across a ball of radius $R^{-u/(2\alpha)}$. Compact support lets every subcritical decay estimate propagate to the gradient, enlarging that radius to $R^{-(u-v)/2}$; as $u, v \to t$, the radius loses no power of R.

3 A lacunary bump bookkeeping lemma

We use one elementary device twice. Fix an even nonnegative $\phi \in C_c^\infty(\mathbb{R}^d)$ with $\int \phi = 1$, put $K = \widehat{\phi}$, and write $e_1 = (1, 0, \dots, 0)$.

Lemma 2 (separated Schwartz bumps). *Let $R_n \to \infty$ be chosen recursively, with $R_{n+1} \ge R_n^2$ and $\log R_n/n \to \infty$. Let $L_n \ge 1$ and suppose $b_n \le C2^{-cn}R_n^{-\beta}$ for some $c > 0$ and $\beta \ge 0$. The radii may be enlarged recursively so that on the disjoint balls*

$$B_n^{\pm} = B(\pm R_n e_1, \rho/L_n)$$

the bump $b_nK(L_n(\xi \mp R_ne_1))$ dominates the sum of all the other bumps by a fixed factor. Here $\rho > 0$ is fixed and small.

Moreover, if $p \ge 1$, $q > -d$, $L \ge 1$, and $LR \ge 2$, then

$$\|K(L(\,\cdot - Re_1))\|_{L^p(|\xi|^q d\xi)} \lesssim_{p,q,K} L^{-d/p}R^{q/p}, \tag{4}$$

$$\|K(L\,\cdot)\|_{L^p(|\xi|^q d\xi)} = C_{p,q,K}L^{-(d+q)/p}. \tag{5}$$

Consequently, upper energy bounds follow by Minkowski's inequality, while each $B_n^{\pm}$ gives the lower contribution

$$\int_{B_n^{\pm}} |F(\xi)|^p|\xi|^q \, d\xi \gtrsim b_n^p L_n^{-d} R_n^q. \tag{6}$$

Proof. Choose ρ so that $|K(\eta) - 1| \leq 1/4$ for $|\eta| \leq \rho$. When R_n is selected, all earlier bumps are Schwartz tails at $R_n e_1$, hence decay faster than any prescribed power of R_n. They can therefore be made smaller than $b_n/16$. The bump at $-R_n e_1$ is handled in the same way. At every later stage, enlarge R_m so that the new tails on all earlier $B_n^{\pm}$, and b_m itself, are bounded by a summable geometric fraction of the corresponding b_n. This proves the domination assertion and also permits the stated lacunarity.

For (4), substitute $y = L(\xi - Re_1)$. On $|y| \leq LR/2$, the weight is comparable to R^q. On the complement, arbitrary Schwartz decay absorbs both the tail and, when $q < 0$, the integrable singularity at $y = -LRe_1$. This gives (4); direct scaling gives (5). Domination on $B_n^{\pm}$, whose volume is comparable to L_n^{-d}, yields (6). $\square$

Only strict inequalities in the critical exponents are used below. Thus the harmless exponential factors in n are dominated by the powers of the superlacunary R_n.

4 The improved compact bound is attained

Theorem 3 (compact sharpness)**.** *For every $d \geq 1$ and every finite $t \geq 0$, there is a compactly supported finite positive measure μ_c on $\mathbb{R}^d$ such that*

$$\dim_{\mathrm{F}} \mu_c = t, \qquad \dim_{\mathrm{F}}^{\theta} \mu_c = t + d\theta \quad (0 \leq \theta \leq 1).$$

Proof. Choose the radii in Lemma 2, set $a_n = 2^{-2n} R_n^{-t/2}$, and define

$$d\mu_c(x) = \phi(x) \left(1 + \sum_{n=1}^{\infty} a_n \cos(2\pi R_n x_1)\right) dx. \tag{7}$$

Since $\sum_n a_n \leq \sum_n 2^{-2n} < 1$, this is a nonnegative, nonzero, compactly supported density. Its Fourier transform is

$$\widehat{\mu}_c(\xi) = K(\xi) + \frac{1}{2} \sum_{n=1}^{\infty} a_n \{K(\xi - R_n e_1) + K(\xi + R_n e_1)\}. \tag{8}$$

For every $v < t$, splitting the sum into centers below, near, and above $|\xi|$, using $R_{n+1} \geq R_n^2$, gives $|\widehat{\mu}_c(\xi)| \lesssim_v (1 + |\xi|)^{-v/2}$. Thus $\dim_{\mathrm{F}} \mu_c \geq t$. On B_n^+, Lemma 2 gives $|\widehat{\mu}_c| \gtrsim a_n$. For every $u > t$,

$$a_n R_n^{u/2} = 2^{-2n} R_n^{(u-t)/2} \longrightarrow \infty,$$

so no u-decay estimate is possible and $\dim_{\mathrm{F}} \mu_c = t$.

Fix $0 < \theta \leq 1$, put $p = 2/\theta$ and $q = s/\theta - d > -d$ (we only need $s > 0$). By (4) and Minkowski, the remote bumps have finite weighted L^p-norm whenever

$$\sum_n a_n R_n^{q/p} < \infty.$$

This holds for every $s < t + d\theta$, because

$$-\frac{t}{2} + \frac{q}{p} = \frac{s - t - d\theta}{2} < 0.$$

The central Schwartz bump is harmless. Conversely, the disjoint peak balls give contributions

$$a_n^p R_n^q = 2^{-2np} R_n^{(s-t)/\theta - d}.$$

For $s > t + d\theta$ these terms tend to infinity, so the energy diverges. Hence the spectrum is exactly $t + d\theta$. $\square$

5 The original Hölder bound is attained without compactness

Theorem 4 (general sharpness)**.** *For every $d \geq 1$, $0 < \alpha \leq 1$, and finite $t \geq 0$, there is a finite positive Borel measure μ on $\mathbb{R}^d$ whose Fourier transform is α-Hölder and for which*

$$\dim_{\mathrm{F}} \mu = t, \qquad \dim_{\mathrm{F}}^{\theta} \mu = t + d\left(1 + \frac{t}{2\alpha}\right)\theta \quad (0 \leq \theta \leq 1).$$

Proof. Set

$$c_n = 2^{-n}, \qquad w_n = 2^{-2n} R_n^{-t/2}, \qquad L_n = (c_n/w_n)^{1/\alpha} = 2^{n/\alpha} R_n^{t/(2\alpha)}.$$

After choosing the radii as in Lemma 2, define positive measures

$$d\mu_n(x) = w_n L_n^{-d} \phi(x/L_n)\{1 + \cos(2\pi R_n x_1)\}\, dx, \qquad \mu = \sum_{n=1}^{\infty} \mu_n. \tag{9}$$

Their total masses are $O(w_n)$, so μ is finite. Furthermore,

$$\int |x|^{\alpha}\, d\mu(x) \lesssim \sum_n w_n L_n^{\alpha} = \sum_n c_n < \infty.$$

The elementary estimate $|\mathrm{e}^{-2\pi i h\cdot x} - 1| \lesssim |h|^{\alpha} |x|^{\alpha}$ therefore shows that $\widehat{\mu}$ is α-Hölder.

The transform is

$$\widehat{\mu}(\xi) = \sum_n w_n \left[K(L_n\xi) + \frac{1}{2}K(L_n(\xi - R_n e_1)) + \frac{1}{2}K(L_n(\xi + R_n e_1))\right]. \tag{10}$$

The same below/near/above-center split used in Theorem 3 gives every decay exponent $v < t$, while the separated peak balls have height comparable to w_n. Since $w_n R_n^{u/2} = 2^{-2n} R_n^{(u-t)/2} \to \infty$ for $u > t$, we obtain $\dim_{\mathrm{F}} \mu = t$.

Again put $p = 2/\theta$ and $q = s/\theta - d$. The central bumps in (10) have weighted norm $O(w_n L_n^{-s/2})$, whose sum is finite. By (4), the sum of the remote-bump norms is finite whenever

$$\sum_n w_n L_n^{-d/p} R_n^{q/p} < \infty.$$

The p-th power of its n-th model term is

$$w_n^p L_n^{-d} R_n^q = 2^{-2np - nd/\alpha} R_n^{-t/\theta - td/(2\alpha) + s/\theta - d}. \tag{11}$$

Thus the energy is finite whenever

$$s < t + d\theta + \frac{td\theta}{2\alpha}.$$

For the reverse inequality, Lemma 2 and the disjoint balls B_n^+ give the lower bound (11). If s is strictly above the displayed threshold, its positive power of R_n dominates the exponential factor and the terms tend to infinity. This proves the asserted exact spectrum. □

6 Answer, checks, and scope

Theorems 1–4 answer both parts of Question 1.4:

- Theorem 1.3 is sharp in the full class of measures with α-Hölder Fourier transform.

- It is not sharp for compactly supported measures with positive Fourier dimension. The exact universal replacement is $\dim_{\mathrm{F}}^{\theta} \mu \leq \dim_{\mathrm{F}} \mu + d\theta$, and compact positive examples attain it simultaneously for all θ.

The symbolic exponent identities in the two constructions and finite numerical truncations of their peak contributions were independently checked by the accompanying script. The proof does not claim to answer the paper's separate questions about coefficient energies, self-similar spectra, or random sumsets. The case $\dim_{\mathrm{F}} \mu = \infty$ is excluded because both upper bounds are then vacuous.

References

[1] J. M. Fraser, *The Fourier spectrum and sumset type problems*, arXiv:2210.07019, Question 1.4 and Theorem 1.3 (source version updated 2026).